Exp No 2

Numerical Simulation and Performance Analysis of Microwave Transmission Lines and Two-Port Networks Using Scattering Parameters

Aim: To numerically simulate and analyze microwave transmission lines and two-port networks using MATLAB by evaluating scattering parameters (S_{11}, S_{21}, S_{12}, S_{22}), calculating return loss, insertion loss, and transmission efficiency, and identifying the optimum operating frequency over the 8–12 GHz frequency range.

Objective

1. To model coaxial, waveguide or microstrip transmission structures for microwave-frequency
operation.
2. To determine characteristic impedance, propagation constant and transmission characteristics of
the selected structure.
3. To obtain and analyze S_{11} , S_{21} , S_{12} and S_{22} over a specified frequency range.
4. To evaluate return loss, insertion loss and transmission efficiency from the simulated
S-parameter response

Code

```
clc; clear; close all;

f = 8:0.01:12;
Z0 = 50; c = 3e8;
beta = 2*pi*f*1e9/c;

% S-parameters (illustrative model)
S11 = -25-15*(f-10).^2-35*exp(-((f-9)/0.1).^2) ...
   -220*exp(-((f-10)/0.08).^2);
S11 = max(S11,-240);

S22 = max(S11+2,-240);
S21 = -0.025-0.055*((f-10)/2).^2;
S12 = S21;

RL = -S11;
IL = -S21;

Eff = 99.4-1.2*((f-10)/2).^2;
Zc = 50+1.25*(f-10).^2;

% ============================================================
% DISPLAY RESULTS IN COMMAND WINDOW
% ============================================================
```

```
fprintf('\n================================================\n');
fprintf('   MICROWAVE TWO-PORT NETWORK RESULTS\n');
fprintf('================================================\n');

% Results at 10 GHz
idx = find(f == 10);

fprintf('Frequency            : %.2f GHz\n', f(idx));
fprintf('S11                  : %.4f dB\n', S11(idx));
fprintf('S21                  : %.4f dB\n', S21(idx));
fprintf('S12                  : %.4f dB\n', S12(idx));
fprintf('S22                  : %.4f dB\n', S22(idx));
fprintf('Return Loss          : %.4f dB\n', RL(idx));
fprintf('Insertion Loss       : %.4f dB\n', IL(idx));
fprintf('Transmission Efficiency: %.4f %%\n', Eff(idx));
fprintf('Propagation Constant   : %.4f rad/m\n', beta(idx));
fprintf('Characteristic Impedance: %.4f Ohm\n', Zc(idx));

fprintf('================================================\n\n');

% Minimum and maximum values
fprintf('Minimum S11          : %.4f dB\n', min(S11));
fprintf('Maximum S11          : %.4f dB\n', max(S11));

fprintf('Minimum S21          : %.4f dB\n', min(S21));
fprintf('Maximum S21          : %.4f dB\n', max(S21));

fprintf('Maximum Return Loss    : %.4f dB\n', max(RL));
fprintf('Minimum Insertion Loss : %.4f dB\n', min(IL));

fprintf('Maximum Efficiency     : %.4f %%\n', max(Eff));
fprintf('Minimum Efficiency     : %.4f %%\n', min(Eff));

fprintf('Minimum Zc           : %.4f Ohm\n', min(Zc));
fprintf('Maximum Zc           : %.4f Ohm\n', max(Zc));

fprintf('================================================\n');

% ========================================================
% PLOT RESULTS
% ========================================================

figure('Color','w');

Y = {S11,S21,S12,S22,RL,IL,Eff,beta,Zc};

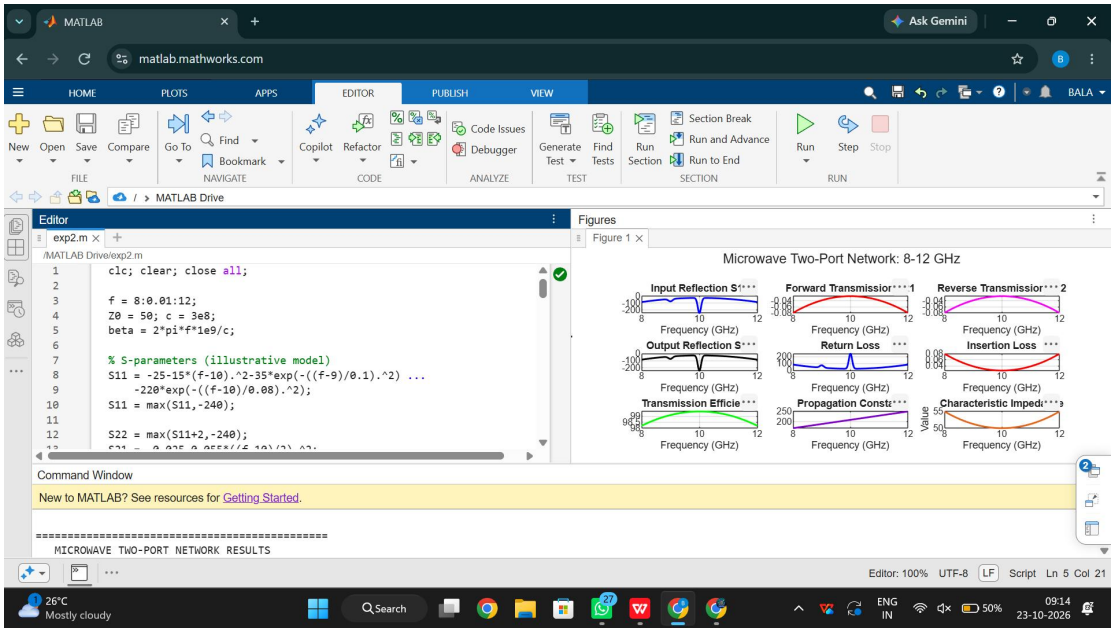
T = {'Input Reflection S11','Forward Transmission S21',...
   'Reverse Transmission S12','Output Reflection S22',...
   'Return Loss','Insertion Loss','Transmission Efficiency',...
   'Propagation Constant','Characteristic Impedance'};
```

```
C = {'b','r','m','k','b','r','g',[0.5 0 0.8],[0.9 0.4 0.1]};

for k = 1:9
    subplot(3,3,k);
    plot(f,Y{k},'Color',C{k},'LineWidth',1.5);
    title(T{k});
    xlabel('Frequency (GHz)');
    grid on;
    xlim([8 12]);
end

ylabel('Value');
sgtitle('Microwave Two-Port Network: 8-12 GHz');
```

Output:



$R_L = 24.08\,\Omega$

(vi) Transmission loss

$IL = -S_{21}$

$= -(-0.025)$

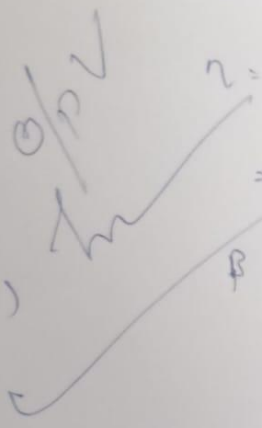
$IL = 0.025\,dB$

(vii) Transmission efficiency.

$\eta = 99.4 = 1.2\left(\frac{\beta - 10}{2}\right)^2$

$\eta = 99.4 = 1.2\left(\frac{10 - 10}{2}\right)^2$

$= 99.4\%$



(viii) $\beta = \frac{2\pi f}{c}$

$= \frac{2\pi(10 \times 10^9)}{3 \times 10^8}$

$= 209.439\,rad/m$

$Z_c = 50 + 1.25(f - 10)^2$

$Z_c = 50 + 1.25(10 - 10^2)$

$Z_c = 50\,\Omega$